General Chemistry I

Chemistry: The central science

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Lectures 3,4: ATOMS

Contents

* Concept of atom: building block of matter

* Evolution of our understanding on atom`s structure

What is matter made of?

Eastern Philosophy



Any matter can be described as:

$$(\mathbf{Fire})_a(\mathbf{Earth})_b(\mathbf{Metal})_c(\mathbf{Water})_d(\mathbf{Wood})_e$$

a-e are continuous and infinite values?

What is matter made of?

2500 years ago: Greek philosophers (Leucippus, Democritus)

Matter cannot be divided indefinitely

It should be composed of very small particles that cannot be divided (small particles = building blocks = Atoms)

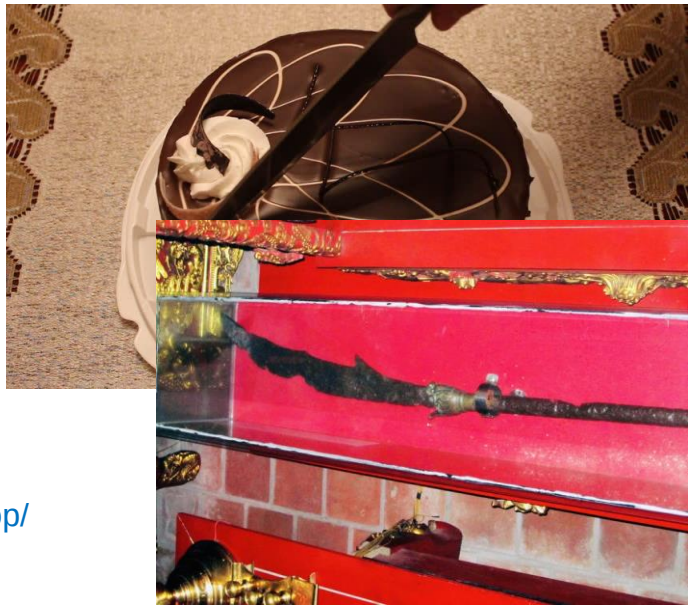
Greek ATOMOS = uncut, indivisible

Lego building blocks are well defined



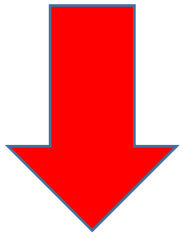
https://seriousplaypro.com/2018/02/11/learned-running-lego-serious-play-workshop/

Cutting a cake in small pieces…
Largely depends on the knife used



18th-19 Century: Wenze, Richter, Proust, Dalton, **Lavoisier**,…

Performed Measurements (Mass, Volume,…)



Matter combines with Matter

in defined quantities

or

multiples of these quantities

John DALTON (1766 -1844)

« A New System of Chemical Philosophy" (1808)



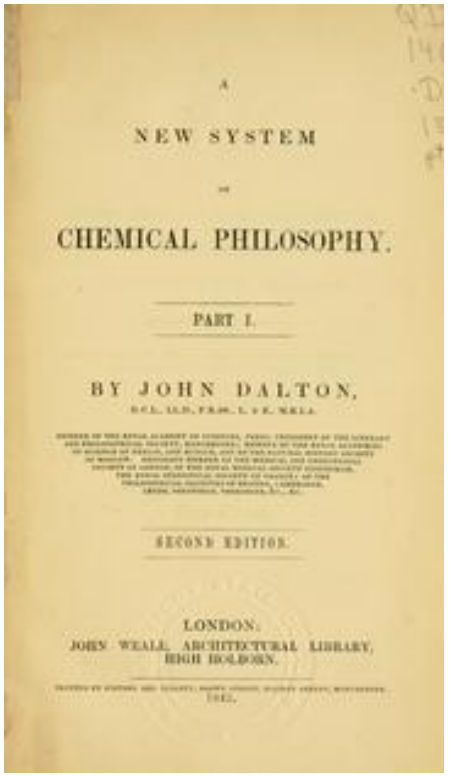
- "Carbonic acid" is formed with 28 parts (% mass) of Carbon and 72 parts (% mass) of Oxygen

- Another gas made of Carbon and Oxygen: contains 28 parts (% mass) of Carbon and 36 parts (% mass) of Oxygen

"Carbonic acid" contains TWICE MORE OXYGEN

(But not 2.3 or 1.8 times)

CO_1 and CO_2 but **NEVER** $CO_{2.3}$ or $CO_{1.8}$



A

NEW SYSTEM

OF

CHEMICAL PHILOSOPHY.

PART I.

BY JOHN DALTON,

SECOND EDITION.

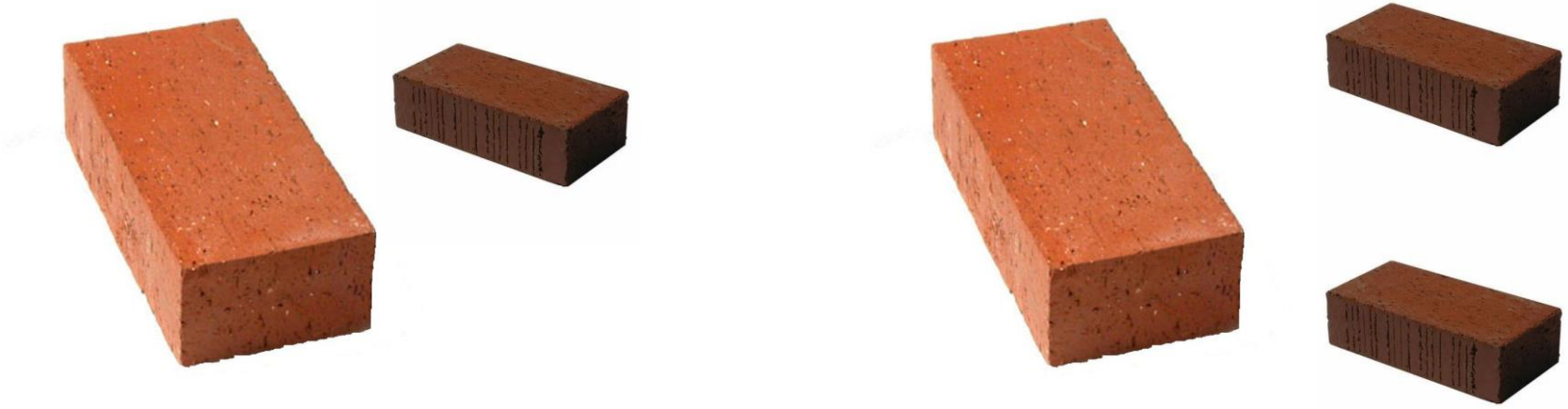
LONDON.

JOHN WEALE, ARCHITECTURAL LIBRARY, HIGH HOLBORN.

And it`s always like this…

"Bricks of matter" combine with "bricks of matter" (ATOMS) in definite proportions

Strong support to Atomic theory



There are 2 laws behind this:

***Law of constant composition*:** In a given compound, the relative numbers and kinds of atoms are constant.

(water is H_2O and never something else...)

Law of multiple proportions*:* If two elements A and B combine to form more than one compound, the masses of B that can combine with a given mass of A are in the ratio of small whole numbers.

e.g. the case of NO_x: N_2O, NO, NO_2

For instance:

A given mass of Carbon will combine with **m mass** of Oxygen or with **2m mass** of Oxygen to yield CO or CO_2

Water H_2O = **16.0 g** of Oxygen + 2.0 g of Hydrogen.

Hydrogen peroxide H_2O_2 = **32.0 g** of Oxygen + 2.0 g of Hydrogen

Ratio of the MASS OF OXYGEN per MASS OF HYDROGEN in the two compounds is **2 : 1**.

BUT…

At Dalton`s time: Impossible to determine chemical formulas

Today we know the two gases are CO_2 and CO

BUT all could be said is that they might be C_xO_{2y} and C_xO_y

Dalton also studies water BUT which formula?

OH, OH_2, HO_2 or something else?

What was missing?

The determination of Mass (or at least Relative Mass) of ATOMS

Or

The determination of the Relative number of ATOMS in a compounds

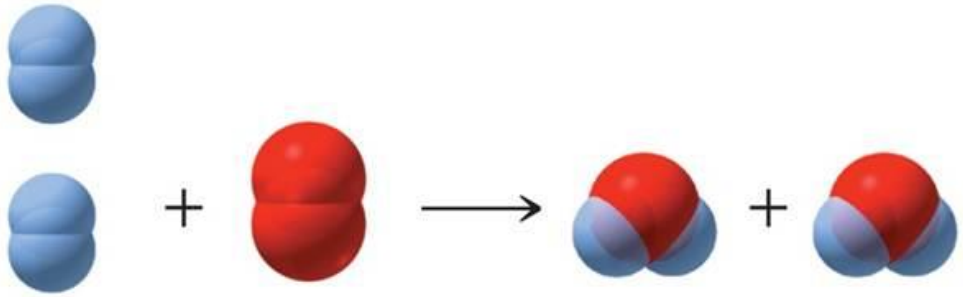
Avogadro (1811): Reactions between gases

Always simple ratios of volumes will react

(2 volumes of Hydrogen and 1 volume of Oxygen reacted together forming 2 volumes of gaseous water) – **Gay Lussac**

In equal volumes of gases, the number of particles are the same

(If pressure and temperature are the same)



$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$$

© 2010 Pearson Education, Inc.

This allows comparing numbers and masses of atoms of different nature and start establish formulas

But it was not so easy…

2 volumes of Hydrogen gas with 1 volume of Oxygen gas = H_2O

Not sure (you must know Hydrogen = H_2 and Oxygen = O_1 … even if, it might be H_4O_2)

Much more difficult work was needed to achieve our current knowledge

Take home messages

Dalton's Atomic Theory

1. Each element is composed of extremely small particles called atoms.

An atom of the element oxygen

An atom of the element nitrogen

2. All atoms of a given element are identical, but the atoms of one element are different from the atoms of all other elements.

Oxygen

Nitrogen

3. Atoms of one element cannot be changed into atoms of a different element by chemical reactions; atoms are neither created nor destroyed in chemical reactions.

Oxygen → Nitrogen



4. Compounds are formed when atoms of more than one element combine; a given compound always has the same relative number and kind of atoms.

N + O → NO

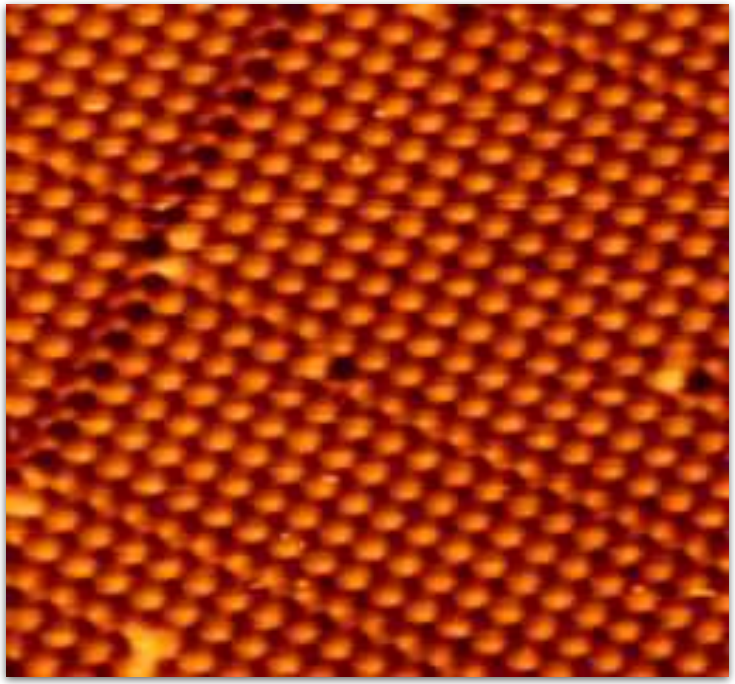
Elements

Compound

The size of Atoms

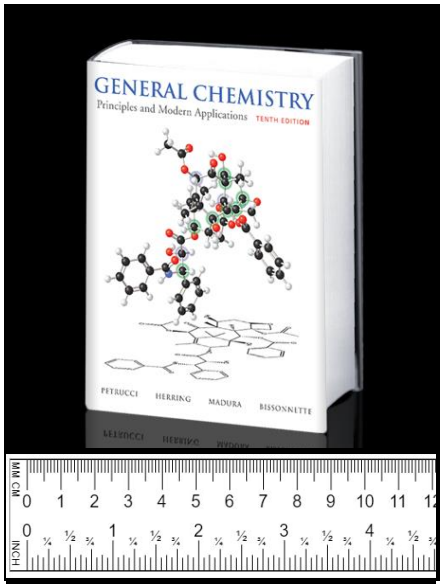
* Most atoms have diameters between 1 x 10^{-10} m and 5 x 10^{-10} m, or 100-500 pm

* Example: Cl atom diameter is 200 picometers (pm) or 2.0Å



Copper atoms

Radius of 1 Cu atom
= 130 pm (pico = 10^{-12})
= 0.13 nm (nano = 10^{-9})
= 1.3 Å (1Å = 10^{-10}m)

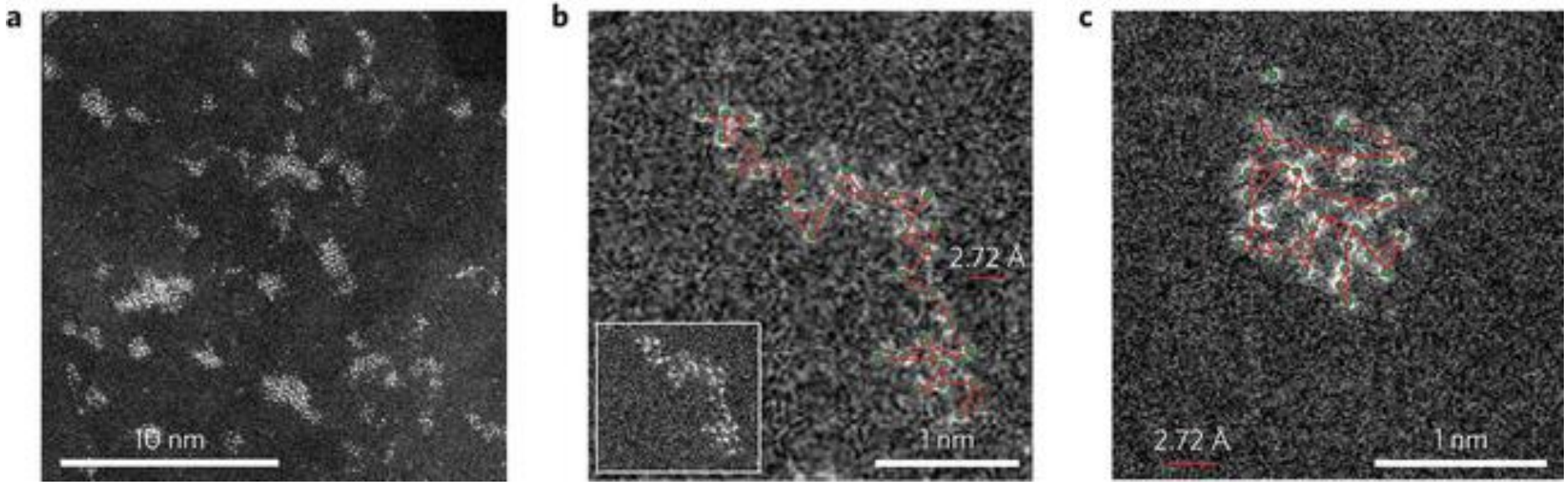


How can we determine the size of an atom (experimentally)?

How can we determine the size of an object?

Structure of *a*-MoS_x material visualized at atomic scale

(by Scanning Transmission Electron Microscopy)



Issue? Money….

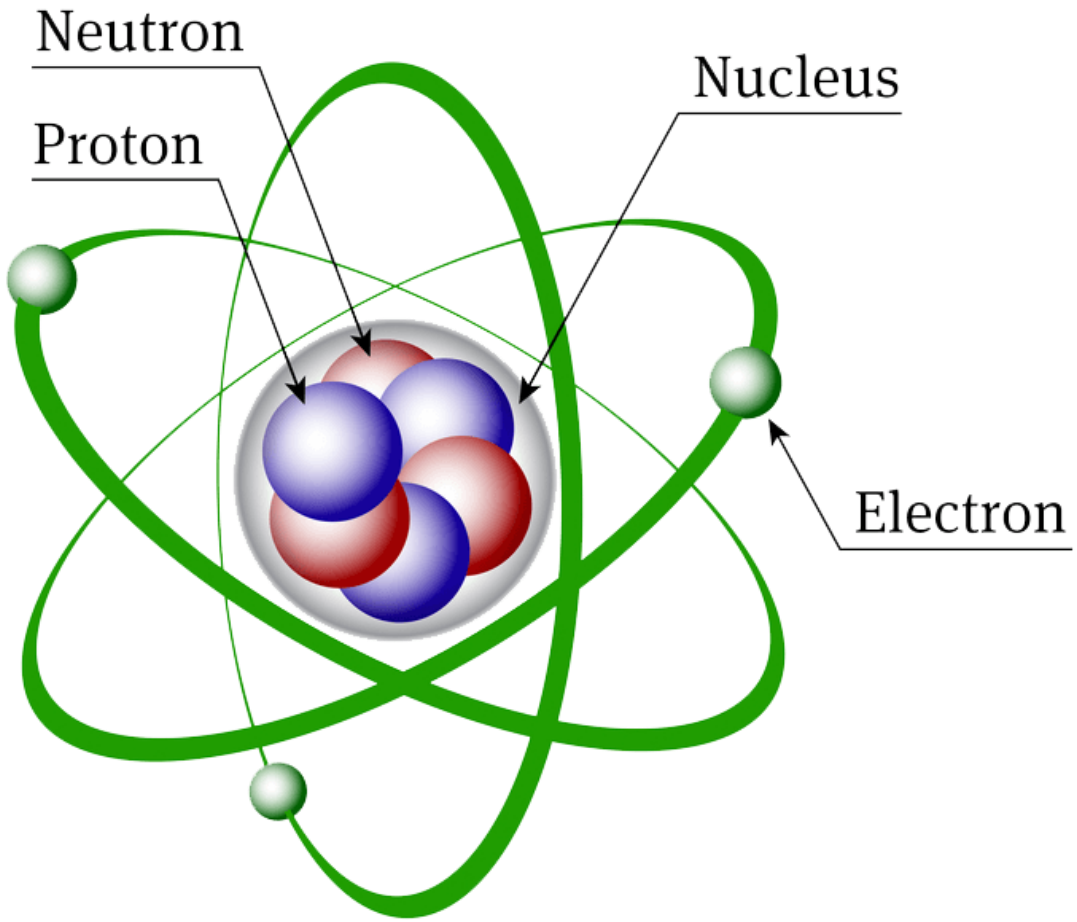
About 1000 USD for 1 measurement in Japan

Tran et al., ***Nature Materials*** **2016**

THE DISCOVERY OF ATOMIC STRUCTURE

What is atomic structure?

THE DISCOVERY OF ATOMIC STRUCTURE



ATOMIC THEORY TIMELINE



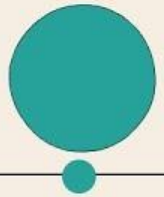
400 BC

Atomos

Democritus



Atoms are the building blocks of matter, the shape of an atom explain elements behavior.



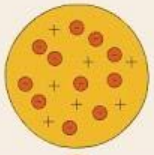
1803

Solid Sphere

John Dalton



Atom is a solid indivisible sphere.



1897

Plum Pudding

J.J. Thomson



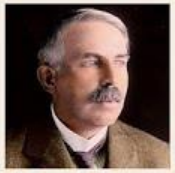
Negative electrons are embedded in a sea of positive charges.



1911

Nuclear

Ernest Rutherford



Positive charges are located within a central nucleus.



1913

Planetary

Niels Bohr



Electrons are restricted in circular orbits with different energy levels.



1926

Quantum

Erwin Schrödinger



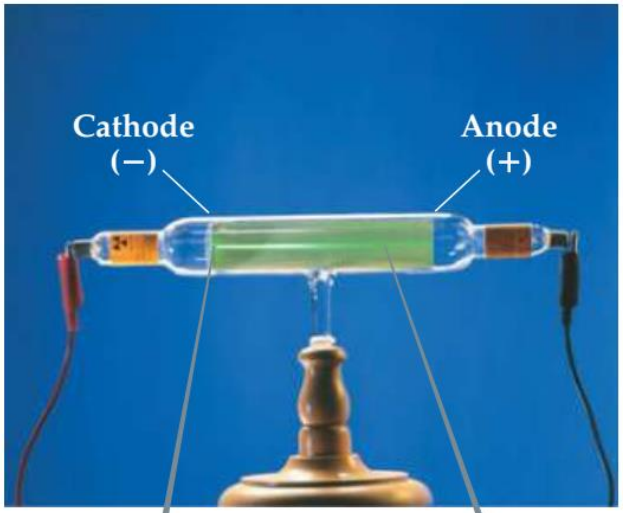
Electrons are in clouds surrounding the nucleus, and this cloud is less dense.

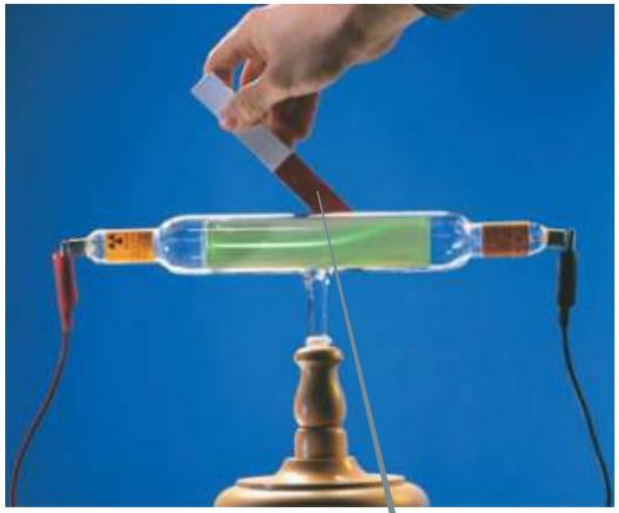
THE DISCOVERY OF ATOMIC STRUCTURE

1- ELECTRONS

THERE ARE MANY ELECTRONS IN MATTER AND THEY ARE RATHER EASY TO EXTRACT

CATHODE-RAY TUBE WITH FLUORESCENT COATING TO SHOW THE PATH OF THE CATHODE RAYS





CATHODE-RAYS ARE EXTRACTED FROM THE NEGATIVE ELECTRODE (CATHODE) AND MOVE TO THE POSITIVE ELECTRODE (ANODE)
NEGATIVE CHARGE

Cathode rays (electrons) move from the negative cathode to the positive anode.

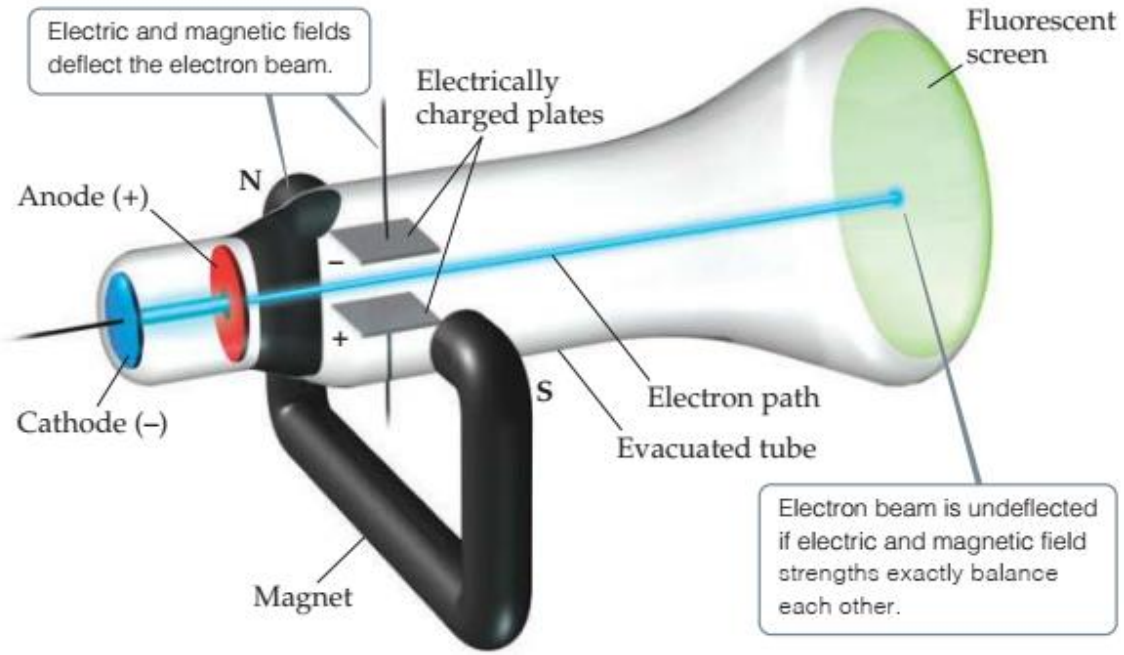
A fluorescent screen is placed in the tube to show the path of the cathode rays. The screen gives off light when a cathode ray strikes it.

The rays are deflected by a magnet.

MAGNETIC AND ELECTRIC FIELDS DEVIATE THE CATHODE RAYS : PARTICLES WITH ELECTRIC CHARGE

https://www.youtube.com/watch?v=vXOeehVTcRA

« CATHODE RAYS ": EXTRACTING ELECTRONS USING AN ELECTRIC FIELD IN A VACUUM (J.J.THOMSON, 1897)



Is the ray a stream of particles or radiation?

1897 : THOMSON : CATHODE RAYS ARE STREAMS OF NEGATIVELY CHARGED PARTICLES later became known as the ELECTRON.

THE CATHODE-RAY TUBE EXPERIMENT ALLOWED HIM TO CALCULATE:

electron charge/electron mass (q/m) = 1.76 10^8 coulombs/gram



CRT display TV

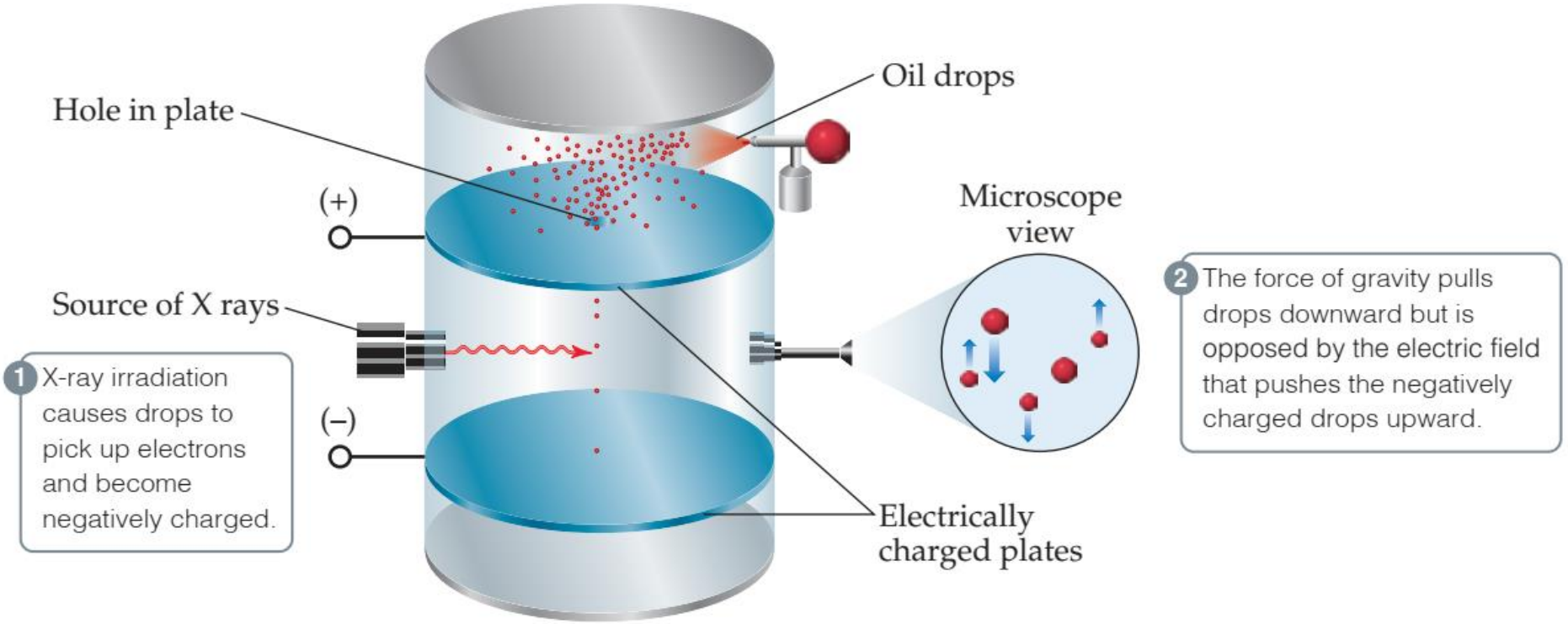


https://www.youtube.com/watch?v=IoJStUt18eo

LED / OLED / QLED TV



Electric charge of electron



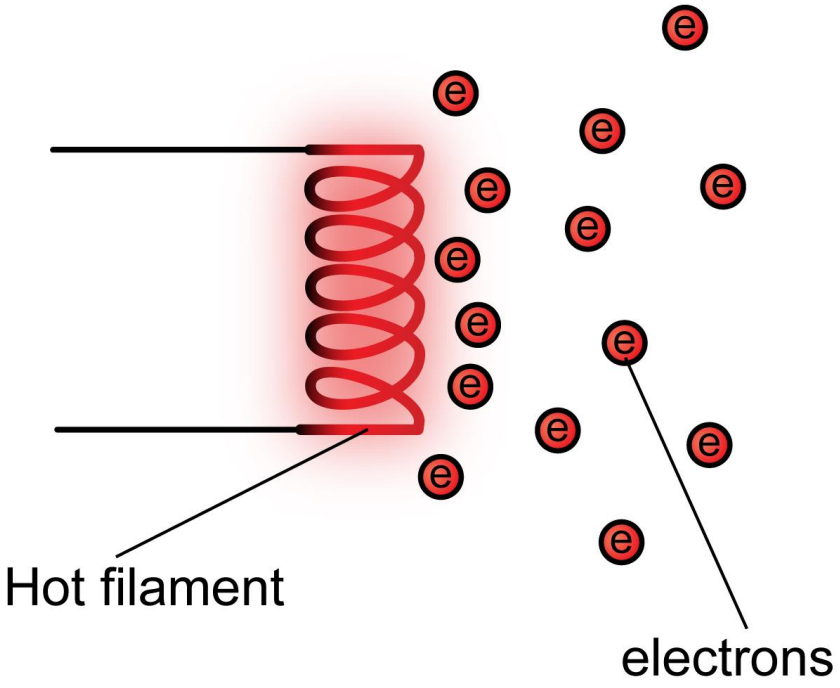
1909 : Robert MILLIKAN : electric charges are always a multiple of a unit charge = that of the electron

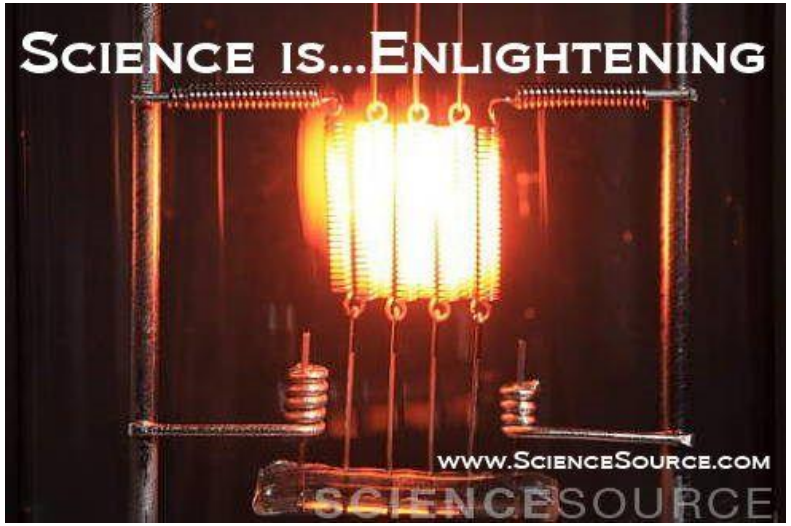
$1e = -\ 1.602\ 10^{-19}$ C

$$\text{Electron mass} = \frac{1.602 \times 10^{-19}\ \text{C}}{1.76 \times 10^{8}\ \text{C/g}} = 9.10 \times 10^{-28}\ \text{g}$$

THERMOELECTRONIC EFFECT (Thermionic emission)

HEATED METAL WIRE EMITS ELECTRONS (ACCELERATED BY ELECTRIC FIELD IN A VACUUM) = (ELECTRON GUN)

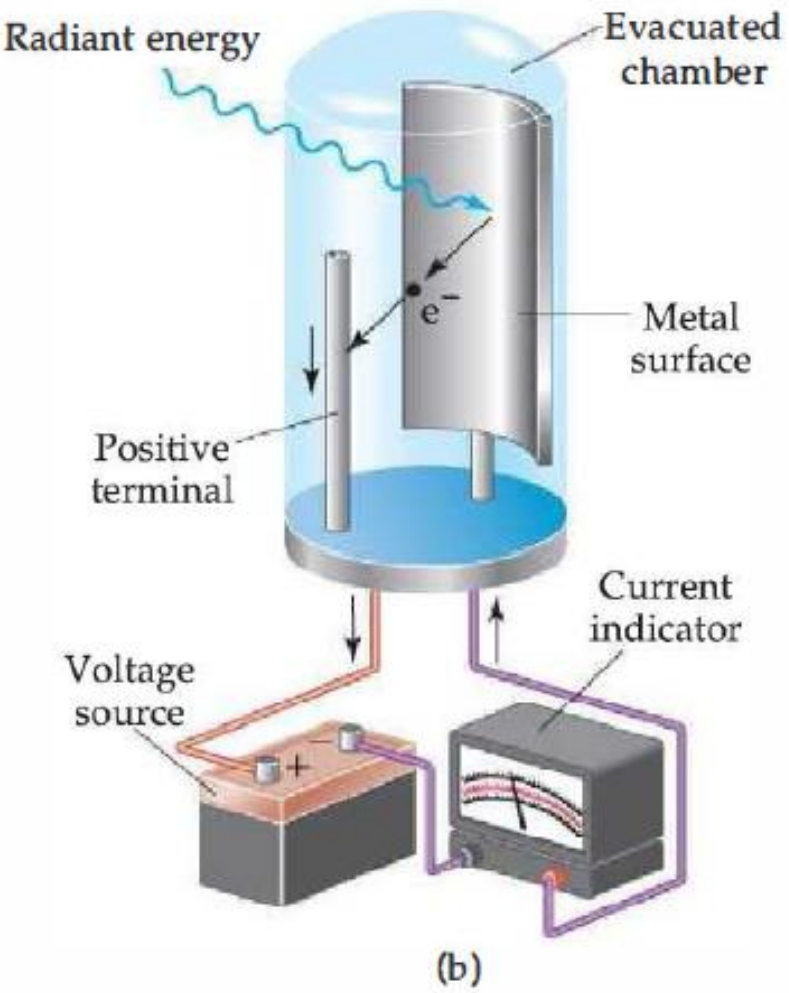




Edison's light bulb

PHOTOELECTRIC EFFECT

EXTRACTING ELECTRONS FROM METALS (OR ANY MATTER) WITH UV OR X-RAYS



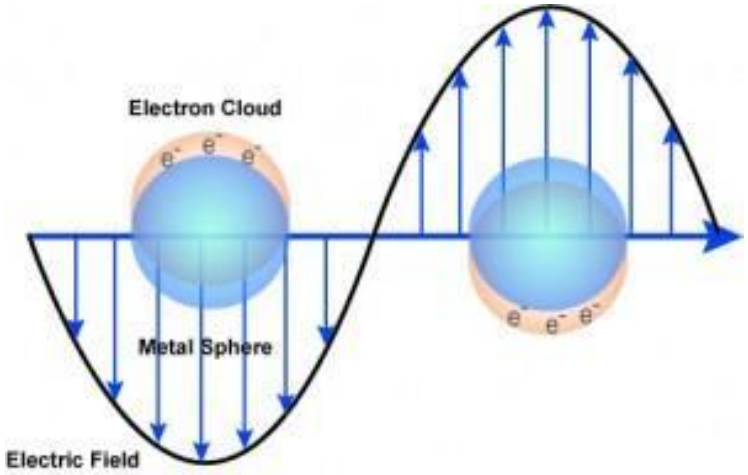
Plasmonic absorption







Michael Faraday and a display of
his colloidal solutions of gold nanoparticles





Colloidal solutions gold NPs having different size

https://nanocomposix.com/pages/plasmonics#target

https://www.indiamart.com/proddetail/colloidal-gold-17658704597.html

SUMMARY : Electron ID card

- **Pointlike** particle (no dimension)
- **Elementary** particle (not further constituant)
- Charge (**negative**) -1.60219×10^{-19} C
- **Very light** (family of leptons): 9.109×10^{-31} kg, 1800 times less than H

Electrons are **numerous** in any matter but their contribution to the total mass is very weak :

In 1 kg of Fe (26 electrons per atom)
the total mass of the electrons is only 550 mg.

WHAT ARE ATOMS MADE OF ?

2.THE NUCLEUS

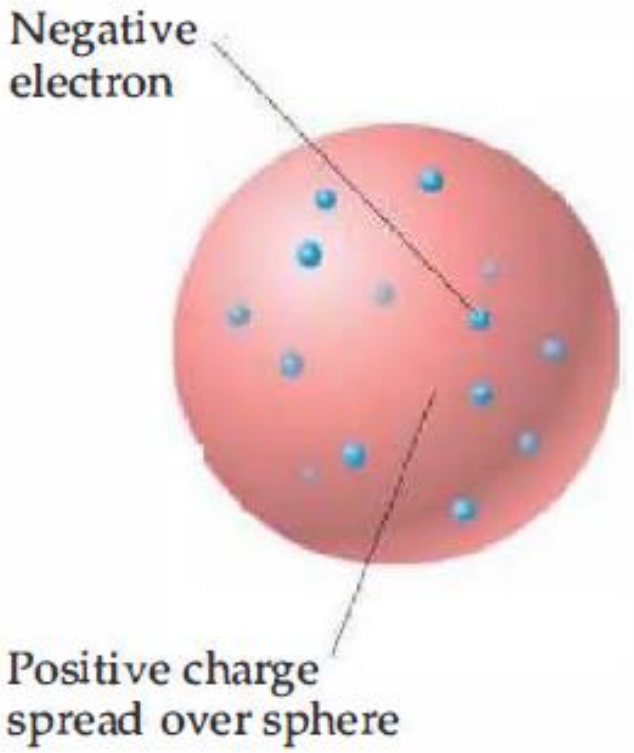
PROBLEM : MATTER IS NEUTRAL

BUT CONTAINS NEGATIVE ELECTRONS ????

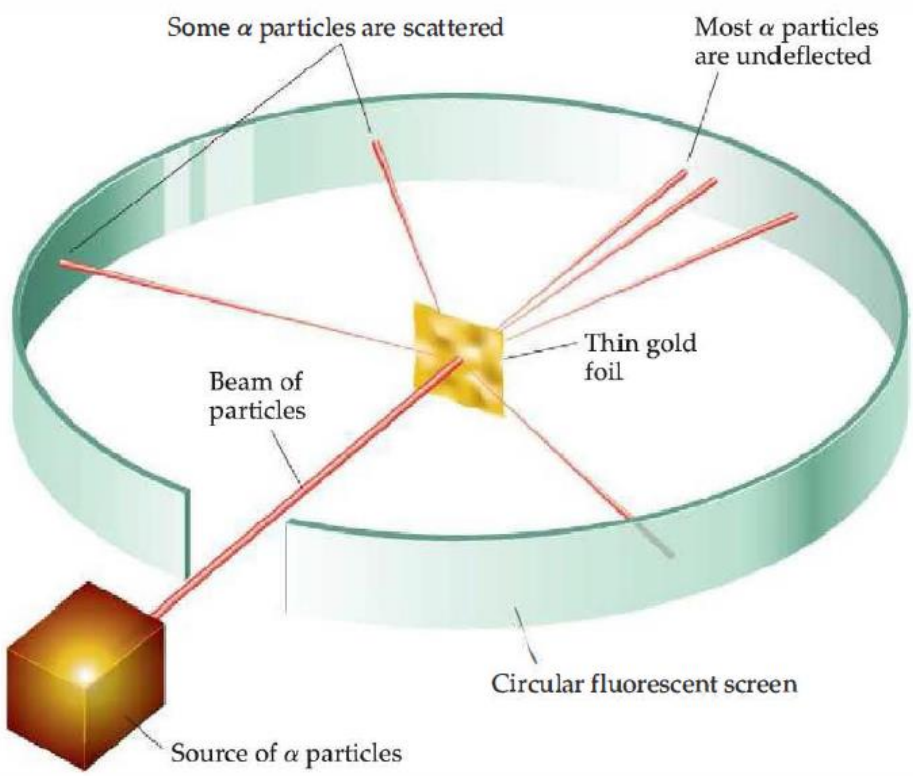
EARLY 1900s : Thomson`s Proposition

ATOM IS A POSITIVE SPHERE OF MATTER

IN WHICH ELECTRONS ARE EMBEDDED



RUTHERFORD (1911) : a decisive experiment



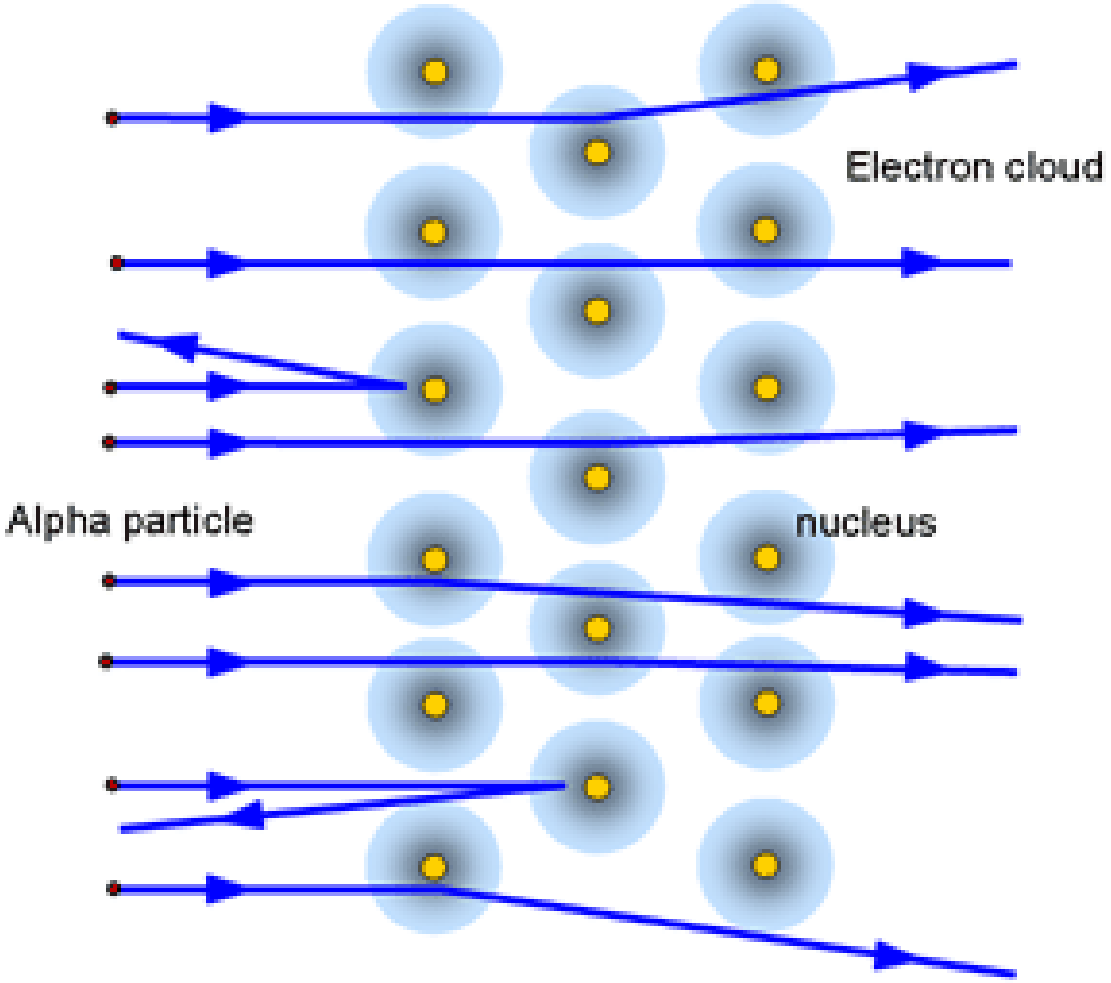
α PARTICLES (POSITIVELY CHARGED) ARE FIRED TO A VERY THIN GOLD FOIL :

UNEXPECTED RESULT !

- **MOST PARTICLES ARE NOT DEFLECTED**

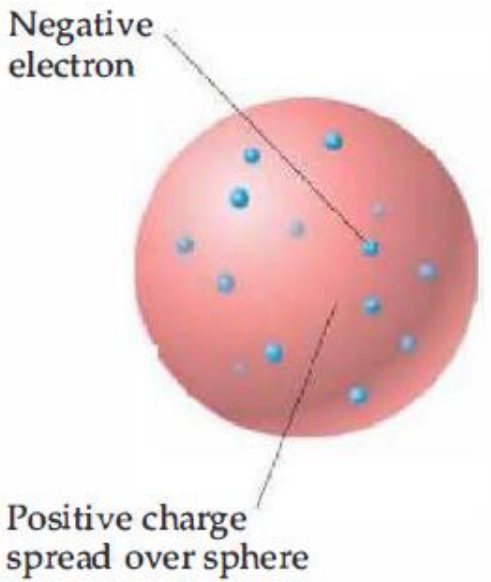
- **SOME OF THEM ARE AND EVEN AT LARGE ANGLES**

Thus, the positive nuclei should be tiny



Thomson`s Proposition:

Not correct



- SO MATTER CONTAINS LARGE EMPTY SPACE !

- BUT EXTREMELY SMALL, DENSE POSITIVE CENTRES OF MATTER DEFLECT SOME α PARTICLES

- SINCE THESE CENTERS ARE VERY TINY, THE PROBABILITY TO MEET THEM IS VERY LOW

- TINY ELECTRONS ARE ORBITING AROUND THESE CENTRES, AT SOME DISTANCE

1919 : RUTHERFORD DISCOVERED PROTON POSITIVE PARTICLES IN THE NUCLEUS

1932 :CHADWICK DISCOVERED NEUTRON NEUTRAL PARTICLES IN THE NUCLEUS.

SUMMARY :

- NUCLEI ARE VERY SMALL AND HEAVY
- MADE OF **PROTONS AND NEUTRONS (NUCLEONS)**

- THESE PARTICLES ARE **NOT ELEMENTARY** (MADE OF QUARKS)

- THEY ARE **NOT POINTLIKE** (DIMENSION $\approx 10^{-15}$ cm).

- THEIR **MASSES ≈ 1800 TIMES THAT OF THE ELECTRON** ARE ALMOST THE SAME

- THE **CHARGE OF THE PROTON IS SYMETRICAL** TO THAT OF THE **ELECTRON**

	PROTON	NEUTRON
MASS (kg)	$1{,}6724\ 10^{-27}$	$1{,}6747\ 10^{-27}$
CHARGE (C)	$1{,}60219\ 10^{-19}$	0

In the very small volume of the nucleus, positive protons repel strongly and the nucleus should explode !!

They are strongly tied together due to another force called strong interaction

FINALLY ! : ATOMS ARE MADE OF

- **very light electrons orbiting at a distance of :**

- **a very tiny nucleus :**
  a few femtometers (fm) = 1.10^{-15}m to 10.10^{-15}m compared to atom (100-500 pm = 1.10^{-10} to 5. 10^{-10} m)

- **but very heavy and dense : most of the mass of matter is in the nucleus**

- **In the neutral atom there are as many electrons as protons**

ELEMENTS and ISOTOPES

IDENTITY OF A NUCLEUS :

ITS NUMBER OF CHARGE = NUMBER OF PROTONS
ITS NUMBER OF MASS = TOTAL NUMBER OF NUCLEONS

Mass =12
Charge = 6

$^{12}_{6}C$

6 protons
6 neutrons

$^{108}_{47}Ag$

Mass =108
Charge = 47

47 protons
61 neutrons

CHEMICAL PROPERTIES OF NEUTRAL ATOMS DEPEND MOSTLY ON THEIR NUMBER OF ELECTRONS = NUMBER OF PROTONS IN THE NUCLEUS

SO CHEMICAL IDENTITY = NUMBER OF PROTONS

THE NUMBER OF NEUTRONS IN THE NUCLEUS MAY VARY SAME CHEMICAL IDENTITY

$^{12}_{6}C$

6 protons
6 neutrons

98.9 %

$^{13}_{6}C$

6 protons
7 neutrons

1.1 %

$^{14}_{6}C$

6 protons
8 neutrons

radioactive

ALL 3 ARE CARBON (VERY SIMILAR CHEMICAL PROPERTIES)

THESE ARE ISOTOPES OF CARBON

THE ELEMENT CARBON IS A MIXTURE OF ISOTOPES

ISOTOPES

$^{12}_{6}C$ — 6 protons, 6 neutrons — **98.9 %**

$^{13}_{6}C$ — 6 protons, 7 neutrons — **1.1 %**

$^{14}_{6}C$ — 6 protons, 8 neutrons — **radioactive**

Isotopes are distinct nuclear species of the same chemical element.

Isotopes have the same atomic number (number of protons, **p**), but different number of neutrons (different **p+n**)

Isotopes of a given element have almost the same chemical properties

BUT they have different atomic masses and physical properties

MORE EXAMPLES

O:	$^{16}_{8}O$: 99,76%	$^{17}_{8}O$: 0,04%	$^{18}_{8}O$: 0,2 %
Cl:	$^{35}_{17}Cl$: 75,4 %	$^{37}_{17}Cl$: 24,6 %	

ATOMIC MASS

DEFINITION :

THE MASS OF 1 ATOM $^{12}_{6}C$ (nucleus+electrons) IS EXACTLY 12 units of atomic mass

THE ATOMIC MASS UNIT (amu) IS 1/12 OF THE MASS OF $^{12}_{6}C$

$1\ amu = 1.66054 \times 10^{-24}\ g$

MASSES OF OTHER ATOMS ARE NOT NECESSARILY INTEGER NUMBERS
(mass of proton ≠ mass of neutron and binding energy of the nucleus changes total mass)

$^{16}_{8}O$: 15.99491 $^{17}_{8}O$: 16.99914 $^{18}_{8}O$: 17.99916.

MASS OF ELEMENTS

SEVERAL ISOTOPES = AVERAGE OF MASSES OF ISOTOPES

C : 12.01 amu (98.9 % of ^{12}C and 1.1% ^{13}C)

Cl : 75.4 % ^{35}Cl and 24,6 % ^{37}Cl with masses 34,96 and 36,96 amu

SO ELEMENT Cl HAS A MASS OF :

(34.96* 75.4 + 36.96 * 24.6)/100 = 35.45 amu

MASS (amu)	
ELECTRON:	0.0005486
PROTON:	1.0073
NEUTRON:	1.0087

AVOGADRO NUMBER, MOLE, MOLAR MASS

AVOGADRO Said : In equal volumes of gases, the number of particels are the same (if pressure and temperature are the same)

BUT He did not know what the number is !

AVOGADRO Number = N = Number of atoms in 12g of ^{12}C
$= 6.022\ 10^{23}$

1 mole = N objects (atoms or molecules)

Mass of 1 mole = mass of N objects (atoms or molecules)

So 1 mole ^{12}C = 12 g

Since 1 amu = 1/12 of the mass of 1 atom ^{12}C,

1 atom ^{12}C has a mass of 12 amu

But one mole of ^{12}C has a mass of 12 g

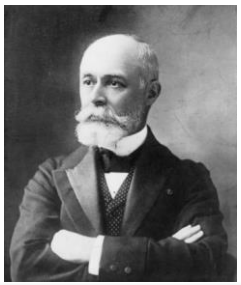
THE SAME NUMBER WILL EXPRESS :
- THE MASS OF 1 ATOM (amu) and
- THE MASS OF 1 MOLE OF THAT ATOM (g)

and :

1 uma = (mass of one atom ^{12}C)/12 = (12/N) /12 grams = 1/N grams = $1/6{,}022\ 10^{23}$ g = $1{,}660\ 10^{-24}$ g.

Not all nuclei are stable

RADIOACTIVITY

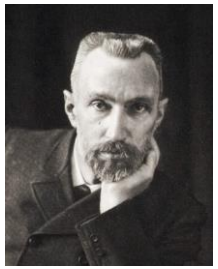


Henri Becquerel

1896

Discovered that Uranium spontaneously emits high-energy radiation

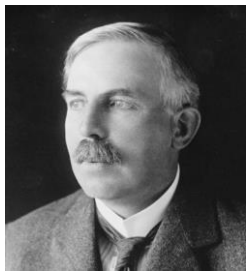




Marie and Pierre Curie

1898

Studied the rays emitted by Uranium, created the term *radioactivity*. Found new elements Polonium and Radium



Ernest Rutherford

1910

Found 3 types of radiations: alpha α, beta β, gamma γ

THE 4 TYPES OF RADIOACTIVITY

α : EMISSION OF A HELIUM NUCLEUS:

$^{A}_{Z}X \rightarrow ^{4}_{2}He + ^{A-4}_{Z-2}Y$ (the nucleus loses 2 protons and 2 neutrons ; changes identity)

β : EMISSION OF AN ELECTRON :

$^{A}_{Z}X \rightarrow ^{0}_{-1}e + ^{A}_{Z+1}Y$ (the nucleus gains one proton : changes identity)

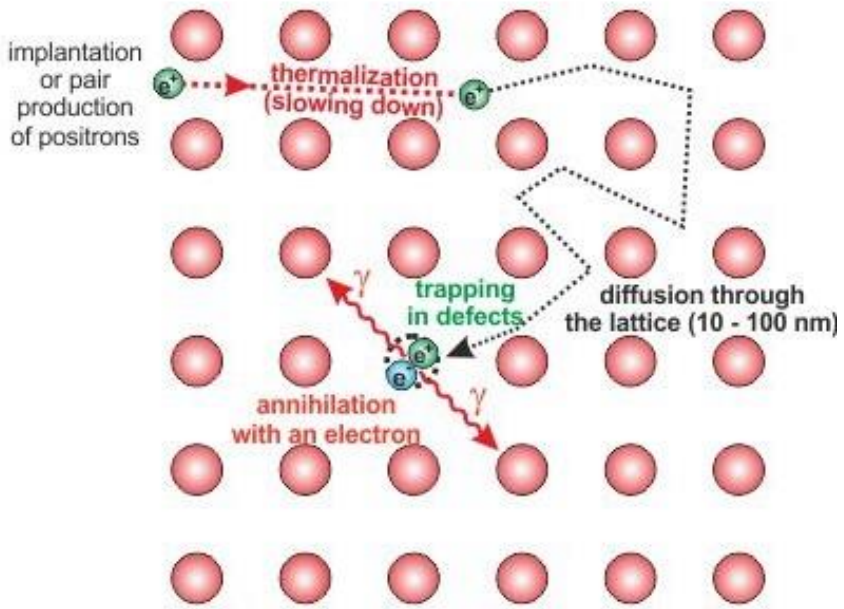
β^{+} : EMISSION OF A POSITON (ANTI-ELECTRON)

$^{A}_{Z}X \rightarrow ^{0}_{1}e + ^{A}_{Z-1}Y$ (the nucleus loses one proton : changes identity)

γ : AN EXCITED NUCLEUS LOSES ENERGY BY EMITTING ELECTROMAGNETIC RADIATION

$^{A}_{Z}X^{*} \rightarrow ^{A}_{Z}X + \gamma$ (the nucleus keeps its identity)

Positron Annihilation Spectroscopy





Dr. Luu Anh Tuyen
(VINATOM)

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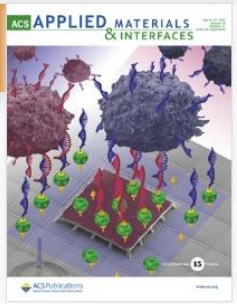
Chuc T. Nguyen, Tuyen Anh Luu*, Thai D. Nguyen, An T. Dam, Ly T. Le, Hyuksu Han, Son T. Lo, Phuc T. Phan, Hue T. Pham, Hue N.T. Nguyen, La Ly Nguyen, Hung Q. Nguyen*, and Phong D. Tran*

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ACS Applied Materials & Interfaces

What is/ are correct information we can provide for the $^{14}_{6}C$ isotope?

a) 6 electron, 6 proton and 8 neutron

b) 6 electron, 8 proton and 6 neutron

c) 6 electron and 14 proton

d) 14 electron and 6 proton

Which of the following pairs of atoms are isotopes of one another?

(a) ^{11}B and ^{11}C

(b) ^{55}Mn and ^{54}Mn

(c) ^{118}Sn and ^{120}Sn

(d) ^{1}H, ^{2}D and ^{3}T

Please identify correct set(s) of isotopes?

a) ^{1}H, D, T

b) ^{12}C, ^{13}C, ^{14}C, ^{14}N

c) ^{58}Fe, ^{58}Co, ^{58}Ni

d) ^{14}N, ^{15}N, ^{16}N

What makes the volume of atom (100-500 pm in diameter)?

a) Size of electron

b) Size of nucleus

c) Total size of electron and nucleus

d) Movement of electron around the nucleus

e) Vibration of nucleus

Let`s consider following species: FeO, $FeO_{1.02}$, Fe_2O_3, and Fe_3O_4. Which of following statements is/ are correct?

a) FeO and $FeO_{1.02}$ are actually identical

b) Fe_3O_4 has both chemical characters of FeO and Fe_2O_3

c) Some of Fe atoms within $FeO_{1.02}$ has similar chemical character to Fe atoms within the Fe_2O_3

General Chemistry I

Chemistry: The central science

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Lectures 3,4: ATOMS